Lambert-Type Closed-Form Approximations for the Double-Exponential Equation

A Generalized Lambert Function Approach via Logarithmic Polynomial Approximation

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Abstract

The nonlinear transcendental equation

$$a + bx + ce^x + de^{e^x} = 0$$

contains both exponential and double-exponential nonlinearities and generally admits no elementary closed-form solution. In this paper we develop a constructive approximation framework based on the substitution

$$x = \ln(1 + y),$$

followed by polynomial approximation of the logarithm. The resulting transformed equations are shown to belong naturally to the Lambert and generalized Lambert families. Explicit approximate solutions are obtained using the classical Lambert-W function and quadratic generalized Lambert functions. Existence, derivation, approximation properties, algorithmic implementation, computational complexity, and interdisciplinary interpretations are discussed.

The paper ends with "The End"

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1 Introduction

Many nonlinear models arising in mathematics, economics, finance, engineering, computer science, machine learning, warfare logistics, population dynamics, strategic planning, astronomy, and systems theory contain nested exponential terms.

A particularly important example is

$$a + bx + ce^x + de^{e^x} = 0.$$

The presence of the double exponential term

$$e^{e^x}$$

typically prevents reduction to standard elementary functions.

The objective of this paper is to obtain constructive closed-form approximations by transforming the equation into forms solvable using Lambert-type functions.

2 The Original Equation

Consider

$$a + bx + ce^x + de^{e^x} = 0. \tag{1}$$

Define

$$x = \ln(1 + y).$$

Then

$$e^x = 1 + y$$

and

$$e^{e^x} = e^{1+y} = e\,e^y.$$

Substituting yields

$$a + b\ln(1+y) + c(1+y) + de\,e^y = 0. \tag{2}$$

Let

$$A = a + c,$$

$$B = c,$$

$$D = de.$$

Then

$$A + b\ln(1+y) + By + De^y = 0. \tag{3}$$

This transformed equation serves as the basis for all subsequent approximations.

3 Linear Logarithmic Approximation

Using the first-order Taylor approximation

$$\ln(1+y) \approx y,$$

we obtain

$$A + (b+B)y + De^y = 0. \tag{4}$$

Define

$$K = b + c.$$

Then

$$Ky + De^y = -A. \tag{5}$$

3.1 Reduction to Lambert-W

Let

$$z = y + \frac{A}{K}.$$

Substitution gives

$$Kz = -De^{-A/K}e^z. \tag{6}$$

Rearranging,

$$ze^{-z} = -\frac{D}{K}e^{-A/K}. \tag{7}$$

Define

$$u = -z.$$

Then

$$ue^u = \frac{D}{K}e^{-A/K}. \tag{8}$$

By definition of Lambert-W,

$$u = W\left(\frac{D}{K}e^{-A/K}\right).$$

Therefore

$$y = -\frac{A}{K} - W\left(\frac{D}{K}e^{-A/K}\right). \tag{9}$$

Substituting the original parameters,

$$y = -\frac{a+c}{b+c} - W\left(\frac{de}{b+c}e^{-(a+c)/(b+c)}\right). \tag{10}$$

3.2 Approximate Solution for x

Since

$$x = \ln(1+y),$$

we obtain

$$x = \ln\left[1 - \frac{a+c}{b+c} - W\left(\frac{de}{b+c}e^{-(a+c)/(b+c)}\right)\right]. \tag{11}$$

4 Quadratic Logarithmic Approximation

Retaining the second-order term gives

$$\ln(1+y) \approx y - \frac{y^2}{2}.$$

The transformed equation becomes

$$A + b\left(y - \frac{y^2}{2}\right) + By + De^y = 0. \tag{12}$$

Hence

$$-\frac{b}{2}y^2 + (b+c)y + (a+c) + De^y = 0. \tag{13}$$

Define

$$P = -\frac{b}{2},$$

$$Q = b + c,$$

$$R = a + c.$$

Then

$$Py^2 + Qy + R + De^y = 0. \tag{14}$$

5 Completion of the Square

Completing the square,

$$P\left(y + \frac{Q}{2P}\right)^2 + \left(R - \frac{Q^2}{4P}\right) + De^y = 0. \tag{15}$$

Let

$$z = y + \frac{Q}{2P}.$$

Then

$$Pz^2 + \Lambda + \Gamma e^z = 0, \tag{16}$$

where

$$\Lambda = R - \frac{Q^2}{4P},$$

and

$$\Gamma = De^{-Q/(2P)}.$$

6 Generalized Lambert Representation

The resulting equation belongs to the generalized Lambert family

$$z^2 + \mu + \nu e^z = 0. \tag{17}$$

Such equations are solved by quadratic generalized Lambert functions. Symbolically,

$$z = W^{(2)}(\mu, \nu). \tag{18}$$

Therefore

$$y = -\frac{Q}{2P} + W^{(2)}\left(\frac{\Lambda}{P}, \frac{\Gamma}{P}\right). \tag{19}$$

Substituting the coefficients yields

$$y = \frac{b+c}{b} + W^{(2)}\left(-\frac{2(a+c)}{b} - \frac{(b+c)^2}{b^2}, -\frac{2de}{b}e^{(b+c)/b}\right). \tag{20}$$

Hence

$$x = \ln\left[1 + \frac{b+c}{b} + W^{(2)}\left(-\frac{2(a+c)}{b} - \frac{(b+c)^2}{b^2}, -\frac{2de}{b}e^{(b+c)/b}\right)\right]. \tag{21}$$

7 Existence Theorem

Theorem 1. *Suppose $b+c \neq 0$ and*

$$1 - \frac{a+c}{b+c} - W\left(\frac{de}{b+c} e^{-(a+c)/(b+c)}\right) > 0.$$

Then the first-order approximation admits a real solution.

Proof. The Lambert-W function is defined whenever

$$z \geq -\frac{1}{e}.$$

Under the stated condition the argument of the logarithm is positive. Hence

$$x = \ln(1+y)$$

is well-defined and real. □

8 Approximation Hierarchy

The methodology generates a hierarchy:

Degree	Approximation	Function Class
1	y	Lambert-W
2	$y - \frac{1}{2}y^2$	Quadratic Lambert
3	Cubic Polynomial	Cubic Lambert
4	Quartic Polynomial	Quartic Lambert
$\vdots$	$\vdots$	$\vdots$
n	Degree-n Polynomial	$W^{(n)}$

9 Algorithm

9.1 Pseudo-Code

```
INPUT: a,b,c,d

Transform:
A = a+c
K = b+c
D = d*e

Compute:
z = (D/K)*exp(-A/K)

w = LambertW(z)

y = -A/K - w

x = log(1+y)

OUTPUT: x
```

10 Computational Complexity

The dominant computational cost is evaluating Lambert-W.

Newton iteration yields

$$O(m)$$

complexity for m iterations.

Thus the overall complexity remains

$$O(m).$$

The method therefore scales efficiently.

11 Applications

11.1 Economics

Nested exponential structures arise in:

- Hyperinflation models.
- Utility functions.
- Recursive expectations.
- Resource exhaustion systems.

11.2 Finance

Applications include:

- Compound risk models.
- Recursive discounting.
- Volatility feedback systems.
- Extreme-event pricing.

11.3 Psychology

Double exponentials appear in:

- Escalating behavioral responses.
- Cognitive overload models.
- Reinforcement cascades.

11.4 Political Science

Examples include:

- Information propagation.
- Political contagion.
- Bureaucratic growth dynamics.

11.5 Geopolitics and International Relations

Applications include:

- Arms-race escalation.
- Alliance cascade models.
- Strategic signaling.

11.6 Warfare Economics

Applications include:

- Mobilization dynamics.
- Supply-chain amplification.
- Attrition escalation models.

11.7 Astronomy

Possible uses include:

- Stellar growth models.
- Recursive radiative feedback.
- Nonlinear gravitational systems.

11.8 Artificial Intelligence

Applications include:

- Deep recursive activations.
- Nested attention structures.
- Multi-agent feedback systems.

12 Discussion

The substitution

$$x = \ln(1 + y)$$

moves the double-exponential equation into a form where logarithmic approximations naturally produce generalized Lambert equations.

The first-order approximation produces an exact Lambert-W representation.

The second-order approximation produces a quadratic generalized Lambert representation.

Higher-order truncations generate an infinite hierarchy of generalized Lambert structures.

13 Conclusion

We have developed a constructive Lambert-based framework for approximating solutions of

$$a + bx + ce^x + de^{e^x} = 0.$$

The transformation

$$x = \ln(1 + y)$$

combined with polynomial approximation of the logarithm yields:

1. A first-order Lambert-W solution.
2. A second-order generalized Lambert solution.
3. A systematic hierarchy of higher-order approximations.
4. Efficient computational algorithms.
5. Broad applicability across mathematics, economics, finance, political science, warfare analysis, astronomy, computer science, data science, and AI.

The resulting framework converts a difficult double-exponential transcendental equation into a sequence of increasingly accurate Lambert-type closed forms.

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Glossary

Lambert-W Inverse of $f(w) = we^w$.

Generalized Lambert Function Extension of Lambert-W solving polynomial-exponential equations.

Double Exponential Expression of the form e^{e^x}.

Logarithmic Approximation Polynomial approximation of $\ln(1 + y)$.

Constructive Solution Explicit algorithm yielding numerical approximations.

Polynomial Truncation Finite-degree approximation of an infinite series.

Recursive Nonlinearity A nonlinearity nested within another nonlinear function.

The End